

Q1.

Alex and Sam are playing a zero-sum game.

The game is represented by the pay-off matrix for Alex.

		Sam		
Alex	Strategy	S₁	S₂	S₃
	A₁	2	2	3
	A₂	0	3	5
	A₃	-1	2	-2

(a) Explain why the value of the game is 2

(3)

(b) Identify the play-safe strategy for each player.

(1)

(Total 4 marks)

Q2.

Victoria and Albert play a zero-sum game. The game is represented by the following pay-off matrix for Victoria.

		Albert			
		Strategy	X	Y	Z
Victoria	P	3	-1	1	
	Q	-2	0	1	
	R	4	-1	-1	

(a) Find the play-safe strategies for each player.

(3)

(b) State, with a reason, the strategy that Albert should never play.

(1)

(c) (i) Determine an optimal mixed strategy for Victoria.

(5)

(ii) Find the value of the game for Victoria.

(1)

(iii) State an assumption that must be made in order that your answer for part

(c)(ii) is the maximum expected pay-off that Victoria can achieve.

(1)

(Total 11 marks)

Q3.

Harry and Will play a zero-sum game. The game is represented by the following pay-off matrix for Harry.

		Will			
Harry	Strategy	D	E	F	G
	A	4	-1	2	3
	B	4	6	3	7
	C	1	3	-2	4

- (a) Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) List any saddle points.

(1)

(Total 5 marks)

Q4.

- (a) Two people, Adam and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Adam.

		Bill		
Adam	Strategy	B₁	B₂	B₃
	A₁	-6	-1	-5
	A₂	5	2	-3
	A₃	-5	4	-4
	A₄	2	1	-4

- (i) Show that this game has a stable solution.

(3)

- (ii) Find the play-safe strategy for each player.

(1)

- (iii) State the value of the game for **Bill**.

(1)

- (b) Roza plays a different zero-sum game against a computer. The game is represented by the following pay-off matrix for Roza.

Computer



	Strategy	C₁	C₂	C₃
	R₁	3	4	-3
Roza	R₂	-2	-1	5

- (i) State which strategy the computer should never play, giving a reason for your answer. (1)
- (ii) Roza chooses strategy R₁ with probability p . Find expressions for the expected gains for Roza when the computer chooses each of its two remaining strategies. (2)
- (iii) Hence find the value of p for which Roza will maximise her expected gains. (2)
- (iv) Find the value of the game for Roza. (1)
- (Total 11 marks)

Q5.

Two people, Rhona and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rhona.

		Colleen		
	Strategy	C₁	C₂	C₃
	R₁	2	6	4
Rhona	R₂	3	-3	-1
	R₃	x	$x + 3$	3

It is given that $x < 2$.

- (a) (i) Write down the three row minima. (1)
- (ii) Show that there is no stable solution. (3)
- (b) Explain why Rhona should never play strategy R₃. (1)
- (c) (i) Find the optimal mixed strategy for Rhona. (7)
- (ii) Find the value of the game. (1)
- (Total 13 marks)

Q6.

- (a) Two people, Tom and Jerry, play a zero-sum game. The game is represented by the following pay-off matrix for Tom.

		Jerry		
Tom	Strategy	A	B	C
	I	-4	5	-3
	II	-3	-2	8
	III	-7	6	-2

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) Rohan and Carla play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Rohan.

		Carla		
Rohan	Strategy	C₁	C₂	C₃
	R₁	3	5	-1
	R₂	1	-2	4

- (i) Find the optimal mixed strategy for Rohan and show that the value of the game is $\frac{3}{2}$.

(7)

- (ii) Carla plays strategy C₁ with probability p , and strategy C₂ with probability q .

Find the values of p and q and hence find the optimal mixed strategy for Carla.

(4)

(Total 15 marks)

Q7.

- (a) Two people, Ann and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Ann.

		Bill		
Ann	Strategy	B₁	B₂	B₃
	A₁	-1	0	-2
	A₂	4	-2	-3

A_3	-4	-5	-3
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Show that this game has a stable solution and state the play-safe strategies for Ann and Bill.

(4)

- (b) Russ and Carlos play a different zero-sum game, which does not have a stable solution. The game is represented by the following pay-off matrix for Russ.

		Carlos			
		<i>Strategy</i>	C₁	C₂	C₃
Russ	R₁	−4	7	−3	
	R₂	2	−1	1	

- (i) Find the optimal mixed strategy for Russ.

(7)

- (ii) Find the value of the game.

(1)

(Total 12 marks)

Q8.

- (a) Two people, Raj and Cal, play a zero-sum game. The game is represented by the following pay-off matrix for Raj.

		Cal			
		Strategy	X	Y	Z
Raj	I	-7	8	-5	
	II	6	2	-1	
	III	-2	4	-3	

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) Ros and Carly play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros, where x is a constant.

Carly		
Strategy	C₁	C₂

Ros	R₁	5	x
	R₂	-2	4

Ros chooses strategy R_1 with probability p .

- (i) Find expressions for the expected gains for Ros when Carly chooses each of the strategies C_1 and C_2 . (2)

- (ii) Given that the value of the game is $\frac{8}{3}$, find the value of p and the value of x . (4)
- (Total 10 marks)

Q9.

Two people, Rowena and Colin, play a zero-sum game.

The game is represented by the following pay-off matrix for Rowena.

		Colin			
		Strategy	C ₁	C ₂	C ₃
Rowena	R ₁	−4	5	4	
	R ₂	2	−3	−1	
	R ₃	−5	4	3	

- (a) Explain what is meant by the term 'zero-sum game'. (1)
- (b) Determine the play-safe strategy for Colin, giving a reason for your answer. (2)
- (c) Explain why Rowena should never play strategy R_3 . (1)
- (d) Find the optimal mixed strategy for Rowena. (7)
- (Total 11 marks)

Q10.

- (a) Two people, Ros and Col, play a zero-sum game. The game is represented by the following pay-off matrix for Ros.

		Col		
	Strategy	X	Y	Z

Ros	I	-4	-3	0
	II	5	-2	2
	III	1	-1	3

- (i) Show that this game has a stable solution. (3)
- (ii) Find the play-safe strategy for each player and state the value of the game. (2)
- (b) Ros and Col play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros.

		Col			
		Strategy	C ₁	C ₂	C ₃
Ros		R ₁	3	2	1
		R ₂	-2	-1	2

- (i) Find the optimal mixed strategy for Ros. (7)
- (ii) Calculate the value of the game. (1)
- (Total 13 marks)**

Q11.

Two people, Rose and Callum, play a zero-sum game. The game is represented by the following pay-off matrix for Rose.

		Callum		
		C ₁	C ₂	C ₃
Rose	R ₁	5	2	-1
	R ₂	-3	-1	5
	R ₃	4	1	-2

- (a) (i) State the play-safe strategy for Rose and give a reason for your answer. (2)
- (ii) Show that there is no stable solution for this game. (2)
- (b) Explain why Rose should never play strategy R₃. (1)

- (c) Rose adopts a mixed strategy, choosing R_1 with probability p and R_2 with probability $1 - p$.
- (i) Find expressions for the expected gain for Rose when Callum chooses each of his three possible strategies. Simplify your expressions. (3)
- (ii) Illustrate graphically these expected gains for $0 \leq p \leq 1$. (2)
- (iii) Hence determine the optimal mixed strategy for Rose. (3)
- (iv) Find the value of the game. (1)
- (Total 14 marks)

Q12.

Sam is playing a computer game in which he is trying to drive a car in different road conditions. He chooses a car and the computer decides the road conditions. The points scored by Sam are shown in the table.

		Road Conditions		
		C_1	C_2	C_3
Sam's Car	S_1	-2	2	4
	S_2	2	4	5
	S_3	5	1	2

Sam is trying to maximise his total points and the computer is trying to stop him.

- (a) Explain why Sam should never choose S_1 and why the computer should not choose C_3 . (2)
- (b) Find the play-safe strategies for the reduced 2 by 2 game for Sam and the computer, and hence show that this game does not have a stable solution. (4)
- (c) Sam uses random numbers to choose S_2 with probability p and S_3 with probability $1 - p$.
- (i) Find expressions for the expected gain for Sam when the computer chooses each of its two remaining strategies. (3)
- (ii) Calculate the value of p for Sam to maximise his total points. (2)
- (iii) Hence find the expected points gain for Sam.

(1)
 (Total 12 marks)

Q13.

Two people, Ann and Bev, play a zero sum game. The game is represented by the following pay-off matrix for Ann.

		Bev			
		Strategy	I	II	III
Ann	I		1	3	6
	II		6	1	1
	III		7	4	2

- (a) Show that there is no stable solution. (3)
- (b) Explain why it will never be optimal for Ann to adopt **Strategy II** and write down the reduced pay-off matrix for Ann. (2)
- (c) Hence, given that the value of the game is $3\frac{3}{5}$, find the optimal mixed strategy for **Bev**. (5)
- (Total 10 marks)
- (You are **not** required to find the optimal mixed strategy for Ann.)